HOLLI
Sustainable Innovation

Method for calculating carbon emissions in Kitsu

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Integration and calculation method

Integrating carbon calculation into a production management tool is relevant because it provides access to a significant amount of information on production.

The calculation of carbon emissions from production in Kitsu is based on :

- ***Emission factors related to the creative industry***
- ***Activity data derived from information recorded by users***

The activity data corresponds to the time units recorded in the schedules:

- ***working hours***
- ***calculation time***

The emission factors correspond to impact models created by HOLLI using its tool Prodo These figures correspond to average values reflecting the activities of animation and post-production professionals. HOLLI is a company specializing in the cultural and creative industries sector, and its approach favors calculations based on physical data rather than monetary equivalents, providing significantly greater accuracy. Emission factors are based on information from databases (see the source data) taken from the reference year 2023.

Carbon emissions correspond to the result of the formula:

Carbon emissions = working hours * temporal emission factor

The impact of a unit of working time is determined by the impact of the means of production, including:

- ***Impact of the dedicated workstation: computer, screens, tablet***
- ***Impact of local technical infrastructure: storage servers, rendering servers, network management***
- ***Impact of cloud usage***
- ***Impact of the building (heating/air conditioning)***
- ***Impact of the meal***

The choice to use working time as the calculation pivot is linked to the highly elastic nature of activity in the animation film and post-production sectors.

Hypotheses

The assumptions are based on data collected from companies in the sector worldwide:

Working day: 8 hours (international average)
5 days a week, no 2x8 or 3x8 shift work

We take a value of 245 for the number of working days based on the fact that on average half of the public holidays fall on working days.

Computer equipment is depreciated over 5 years.
The depreciation of other equipment is done over 10 years.
Depreciation is calculated based on hours worked (8 hours/day).
The manufacturing of buildings is not taken into account in the model at the moment, as this value is pending consensus on carbon calculation.

Impact of the meal

The meal impacts calculation uses an impact study
https://nutritionj.biomedcentral.com/articles/10.1186/s12937-021-00669-6/tables/3
We used the study countries as proxies for their respective regions; for example, the data for Thailand represent the data for Asia. The assumption is that the regions use the factors of the countries they include, so the data for Africa and Oceania correspond to European values.We have added ADEME data for metropolitan France

The limitations of this approach:
The figures correspond to the recommendations for a balanced meal (2000kcal/day).
This assumption allows for relatively good accuracy values in Western countries and presents a level of accuracy equivalent to methods using global values when available.

Impact of the dedicated workstation

All the equipment taken into account for the workstation:

- A table, a chair (source: ADEME)
- A Cintiq-type graphics tablet (source: Boavizta)
- 1 écrans DELL 24' (source DELL p2425H)
- Station DELL Precision 5860 32 GB RAM GPU NVidia RTX A1000
- 1/40 of a NAS modeled on an APY AMD EPYC server base with 12 SSDs
- 1/40 d'un Switch 48 ports 10GbE Netgear

This is an average value for the graphics functions of a 2D or 3D studio. In reality, this configuration varies depending on the roles, the number and power of the equipment, which affects electricity consumption by approximately 30%. The machine is installed in the workspace while storage and rendering are located in the server room, which is the most common practice.

Impact of local technical infrastructure

The setup varies considerably from one studio to another, particularly depending on whether the work is 2D or 3D, as 2D studios have limited rendering infrastructure. The calculation of the rendering impact is separate from the operator calculations and retrieves production-specific information. The air conditioning cooling capacity is calculated to be equivalent to the power used.
We consider the equivalent of a Netgear 48-port 10GbE switch consuming an average of 120W under heavy usage. We take into account 1/40 of its impact. We also consider the impact of the air conditioning used to cool this equipment.

The storage server is based on an AMD EPYC 9004 series FULL NVMe 295TB NAS configuration (368TB RAW), the impact of which is modeled onhttps://datavizta.boavizta.org/serversimpact

The electricity consumption of the entire workstation, equipment and office, including the air conditioning portion corresponding to the total electricity consumption, which takes into account a distribution between different levels of use (office work, business tools, rendering), is 409W.

Impact of cloud infrastructure

Ccloud usage is limited to data transfers between the studio and office applications (email, instant messaging, document management), as well as project management with tools like Kitsu. It is also very common for artists to look at video tutorials and listen to streaming music.
If the entire studio infrastructure is in the cloud, the impacts will be slightly greater but of the same order of magnitude as on-premise equipment.

We consider professional uses to have an equivalent impact to personal uses. This information comes from an article published in the scientific journal Nature comparing the impacts of digital uses in different countries.The environmental sustainability of digital content consumption | Nature Communications

Impact of the building

The following are taken into account:

- The impact of office heating
- The impact of air conditioning for office and technical room, impact of electricity consumption and leakage emissions

The impact is calculated based on the time and average surface area of an operator (5m2).

The impacts of heating and air conditioning offices are calculated in relation to a reference temperature value and the variation of this value during the year in the country concerned, which modulates a correction factor for related energy consumption.

Heating is active during heating season, i.e., months when the average temperature is below 13°C, according to data from the Copernicus Climate Change Service (2025) available at OurWorldinData.org/climate-change. It is based on a value of 300W per workstation, or 60W/m² (EIA value) multiplied by 5m², the average size of a graphic designer's workspace in an open-plan office. This value can vary significantly, from one country to another, but no reference data is available to our knowledge.

Office air conditioning is active during months when the average temperature is above 21°C, according to data from the Copernicus Climate Change Service information (2025) available onOurWorldinData.org/climate-change

In the model we use an air conditioner charged with 0.25 kg of R22 per Watt of cooling produced; these values are derived from documentary research on practices:

- UNEP Residential and Commercial Air Conditioning and Heating
- https://archive.ipcc.ch/pdf/special-reports/sroc/sroc05.pdf
- https://bregroup.com/documents/d/bre-group/b_airconditioningenergyuseannexbfinal
- https://climate.ec.europa.eu/document/download/64c45c9d-de32-4bba-8eae-423cf74a6073_en?filename=2020_03_25_hfc_alternatives_en.pdf

We consider a fluid loss of 10% per year, which is the default value in the Carbon Footprint methodology.

The COP (Coefficient of Performance) of an air conditioner is an indicator that measures the ratio between the thermal energy produced (heating or cooling) and the electrical energy consumed by the unit. The value chosen is 4, which is an average value for air conditioners.https://fr.wikipedia.org/wiki/Coefficient_de_performance

Rendering configuration

The rendering machine is based on a dual-processor configuration.2x16 core + 2 * 128Go RAM + 1 SSD 1To + // alim 850Wwhose impact is modeled onhttps://datavizta.boavizta.org/serversimpact

We consider the equivalent of a Netgear 48-port 10GbE switch consuming an average of 120W under heavy usage. We take into account 1/40 of its impact

The storage server is based on an APY Micro configuration: APY STG12 AMD EPYC 9004 series FULL NVMe 295TB (368TB RAW), the impact of which is modeled onhttps://datavizta.boavizta.org/serversimpact

The weighted usage pattern (standby/normal use/full power usage) for storage results in an average consumption of 142W. The total consumption for the rendering server plus a portion of the storage and network is 775W.

Elements not taken into account

- The scope covered is solely that overseen by Kitsu, and therefore essentially the manufacturing phases: pre-production, distribution, and promotion are not taken into account.
- Service emissions, corresponding to the climate impact of support services (administration, production, technical support), are not taken into account in this calculation. They represent between 10 and 20% of additional emissions depending on the company's structure and location.
- The means of transportation between home and work vary greatly, as the values are both extremely variable (ranging from a few dozen grams in Paris to several kilograms in Los Angeles) and cannot be localized by country because the

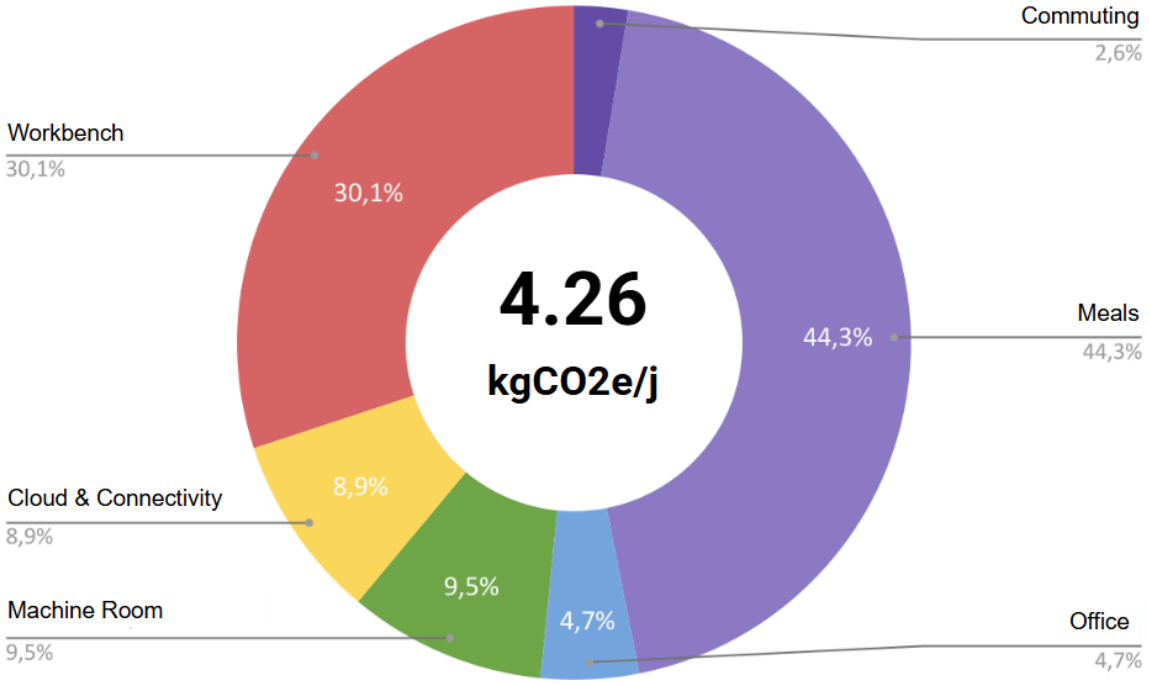
differences stem primarily from urbanization characteristics. This information could be requested from the user in a future version.

- The impacts of building construction and works
- The manufacturing impacts of heating and air conditioning systems
- The means of transport used to visit the different studios in the case of international co-productions.
- Differences in heating and air conditioning technology: the differences are significant, but a lot of information would need to be requested from the user.
- The differences in thermal insulation between buildings again show significant variability, as energy consumption can vary from one to three times the same temperature difference for both heating and cooling, but the data is difficult to integrate.
- The differences in impact related to production pipelines can be significant, especially compared to the impacts related to storage and rendering. However, since we consider both processing time and rendering time, this difference is minimized, given that the difference in network and storage impacts is minimal.
- The impacts directly related to AI, partly included in cloud usage, however ongoing assessments show a much greater impact than traditional web usage.

Overall error level is better than that of a report based on monetary factors because it relies on data much closer to the reality of companies in the sector. It is quite comparable to the difference of LCAs of similar products made by different manufacturers.

Orders of magnitude

Here are the values obtained for a Parisian VFX studio, for a **total of 4.26 kgCO2e per working day** for a graphic designer.



Source data

The data used comes from the following sources:

- ADEME Footprint Database : building(furniture, weight factor, IT products)
- Agribalyse: meals (FR)
- Negaoctet: factors integrated into the Ademe database for IT services
- Boavista: modeling the impact of computer hardware
- Hardware manufacturer data (DELL, WACOM…)
- Data from independent studies:
    - https://nutritionj.biomedcentral.com/articles/10.1186/s12937-021-00669-6/tables/3
    - CNRS and German Ministry for the Environment Renater project for cloud impacts
    - The environmental sustainability of digital content consumption | Nature Communications
    - OurWorldinData.org/climate-change

Data sources are standardized to standard time frames: year, day, hour. Depending on the country, the number of working days and days worked varies.